Zoom In, Scale Out

Metric depth model gulo ki digital zoom er physics mane?

Research overview · Rabeya Noor · target CVPR 2027 · 19 October 2026

Ek line e: Chhobi crop kore boro korle (digital zoom) focal length thik s gun bare, kintu kono jinish kachhe ashe na. Amra dekhiyechi beshirbhag metric depth model zoom korle depth kom bole: shobcheye popular Depth Anything V2 3x zoom e depth prai 1/3 bole (α = 0.95). Ete label lage na, theory ache, ar label chhara fine-tuning (EFT+B) diye eta prai puro thik kora jay.

1. Shomossha

Self-driving, robot, AR ar phone camera te ekta chhobi theke protita jinish koto **meter** dure, sheta bolte metric depth model lage (Depth Anything, Depth Pro, MoGe, UniDepth, Metric3D, Depth Anything 3). Meter e bolte hole camera er focal length jana dorkar. Karon wide lens e kachher jinish ar telephoto lens e durer jinish ekdom ek rokom dekhate pare. Model gulo camera chhara shudhu pixel dekhe eta anuman kore. Proshno holo: ora ki asholei camera bujhe, naki shudhu "boro dekhale kachhe" shortcut use kore?

2. Mul idea: digital zoom ekta free, shothik ruler

Chhobir ekta box crop kore abar ager size e resize korle (digital zoom), chhobi ta ekdom shothik bhabe ekta notun pinhole camera theke tola chhobir moto hoy: f' = s·f, principal point shore jay, kintu **depth bodlay na**. Tai label chhara test kora jay: zoom er age ar pore model nijer depth ar focal er sathe tulona korlei hoy.

Shonkhya	Mane	Thik model	Shortcut model
α (depth exponent)	D̃(zoom) / D̃(original) ≈ s^−α	0	1 (2x zoom e depth ordhek)
β (focal exponent)	f̃(zoom) / f̃(original) ≈ s^β	1	0 (focal bodlay na)

3. Theory

3.1 Tinta invariant

- α = 0 → depth thik thake;
- β = 1 → 3D shape thik thake;
- α + β = 1 → shudhu pasher (lateral) map thik thake, depth ba shape na. "Boro dekhale kachhe" reasoning thik ei line e pore: Ḋ = f̃·L/ℓ hole zoom e Ḋ ∝ s^(β−1).

3.2 Zoom–squash ambiguity (notun)

Focal s·f e tola chhobi pixel-by-pixel sheyi scene er chhobi, jeta depth er dik borabor s gun chepta kora: u = s·f·X/Z = f·X/(Z/s). Chepta scene e jinish er width ek thake, shudhu gobhirota bhul hoy (4.5 m er gari 2x e 2.25 m lomba hoye jay). Tai shudhu size cue diye zoom dhora oshombhob, 3D shape prior lage. Ar shoja dewal (fronto-parallel plane) e kono model-i camera na jene zoom dhorte parbe na.

3.3 Stale K

Metric3D er moto camera-input model e zoom er por purono K dile network ekoi pixel dekhe, shudhu shesh e scale bodlay. Tai α(stale) = α(true) + 1, eta ganit diye proman kora jay. Data te median tofat 1.000.

4. Experiment setup

Ki	Details
Model (main)	DA-V2-S/B, ZoeDepth (pixels only) · Depth Pro, MoGe-2-S, UniDepthV2-S (predicts camera) · Metric3D v2-S, thik K ar stale K (takes camera)
Model ladder	DA-V2-L, MoGe-2-B/L, UniDepthV2-B/L, UniK3D-L, Metric3D v2-L, Depth Anything 3 (Nested-Giant, Metric-L, L), VGGT-1B
Data	KITTI val selection (LiDAR), NYUv2 test (Kinect), DIODE val indoor + outdoor (laser). Protita theke 200 chhobi, shob model e ekoi chhobi (800 chhobi/model)
Zoom	1, 1.25, 1.5, 2, 3x majhkhane · off-centre: 4 kona, 50 chhobi (KITTI, NYU)

Ki	Details
Control	blur control (zoom er moto jhapsa, field of view ek) · geometry check (pixel error 0) · unzoomed prediction ekoi pixel e (fair reference) · result verify tool (56 file, shob check pass)
Statistics	95% bootstrap CI · paired Wilcoxon signed-rank + Holm correction · Spearman with bootstrap CI
Compute	Kaggle T4 GPU; laptop e shudhu analysis

5. Result 1: zoom audit

Model	Group	α [95% CI]	β	α+β	Blur control	AbsRel 1x	AbsRel 3x
DA-V2-S	pixels only	0.95 [0.93, 0.99]	–	–	1.3%	0.317	0.652
DA-V2-B	pixels only	0.85 [0.82, 0.87]	–	–	0.8%	0.313	0.599
ZoeDepth	pixels only	0.61 [0.58, 0.64]	–	–	0.7%	0.310	0.539
Depth Pro	predicts camera	0.31 [0.27, 0.34]	0.65	0.97	2.8%	0.310	0.459
MoGe-2-S	predicts camera	0.36 [0.34, 0.38]	0.56	0.94	3.1%	0.274	0.465
UniDepthV2-S	predicts camera	0.34 [0.30, 0.40]	0.45	0.87	4.0%	0.824	0.792
Metric3D v2-S	takes camera (true K)	-0.07 [-0.08, -0.06]	–	–	1.6%	0.142	0.358
Metric3D v2-S (stale K)	takes camera (stale K)	0.93 [0.92, 0.94]	–	–	1.6%	0.142	0.628

Note: ei table er AbsRel 4 ta dataset er gor (DIODE-outdoor er boro error shoho). Section 7 er EFT table e protita dataset alada, ar sheta 392 px input e chale. Tai DA-V2-S er 1x AbsRel ekhane 0.317, ar EFT table e KITTI te 0.138. KITTI te default preprocessing e eta 0.133, mane duita mile jay. CI ekhane image-level; paper e scene-level cluster bootstrap use kora hoy, jeta aro chouro.

Ja pelam:

- Pixels-only model gulo prai puro shortcut use kore: DA-V2-S α = 0.95.
- Camera predict kora model gulo ordhek bujhe: α ≈ 0.31–0.36, ar focal 3x er jaygay 1.63–2.04x bole.
- Depth Pro e protita chhobite α+β er median 0.97, mane theory er "lateral invariant" line e prai hubohu thake.
- Thik K pele Metric3D prai equivariant (α = -0.07), stale K dile pixels-only model er moto (0.93).
- Blur control e depth bodlay mattro 0.7–4%, tai dosh interpolation er na, camera bojhar.

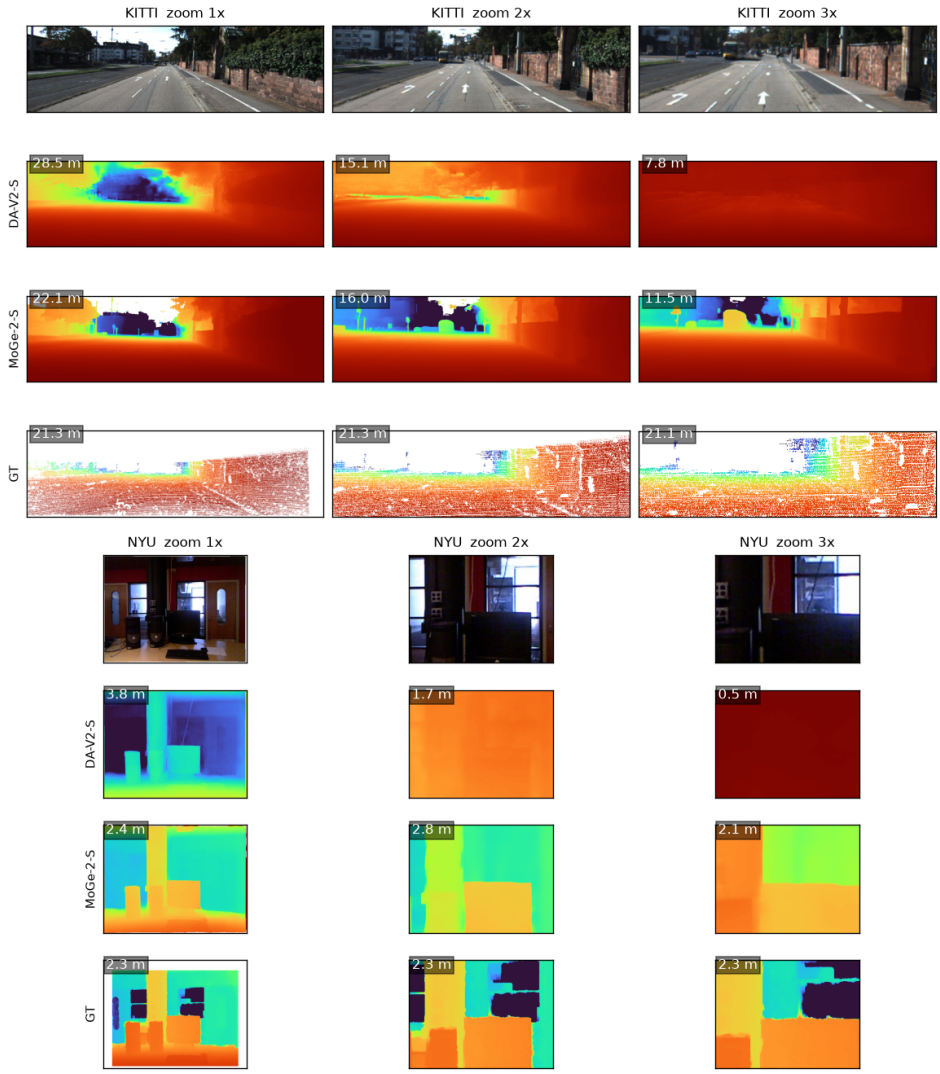


Figure 1. Ekoi scene 1x, 2x, 3x zoom e. Ek row er shob depth panel ekoi rong er scale e (meter). DA-V2-S zoom korle "kachhe" chole jay (KITTI 28.5 → 7.8 m, GT 21 m); MoGe-2 ordhek.

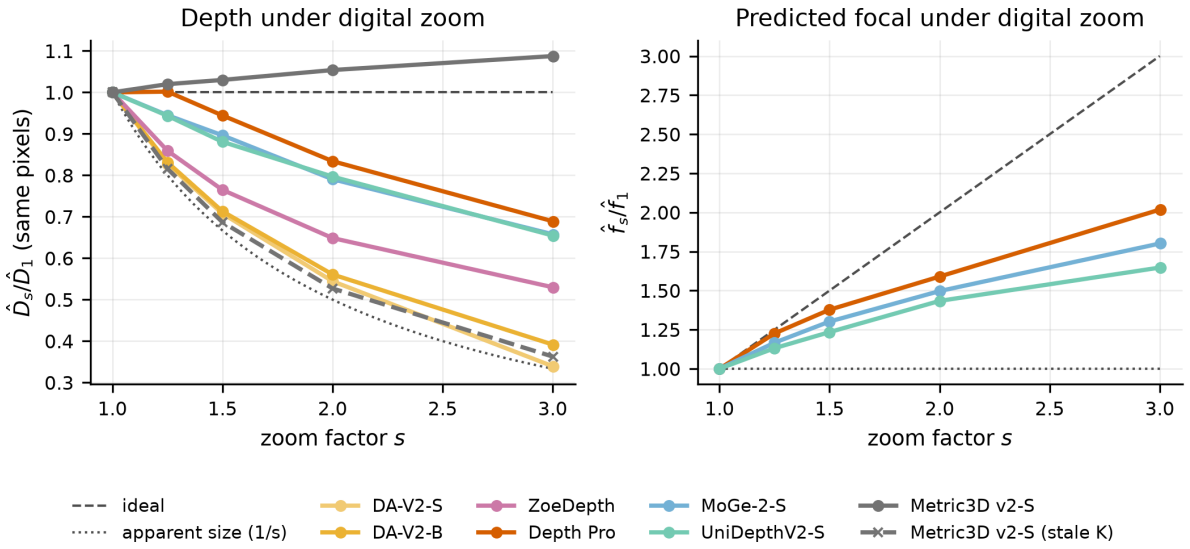


Figure 2. Zoom e depth (bame) ar predicted focal (dane), 800 chhobir median.

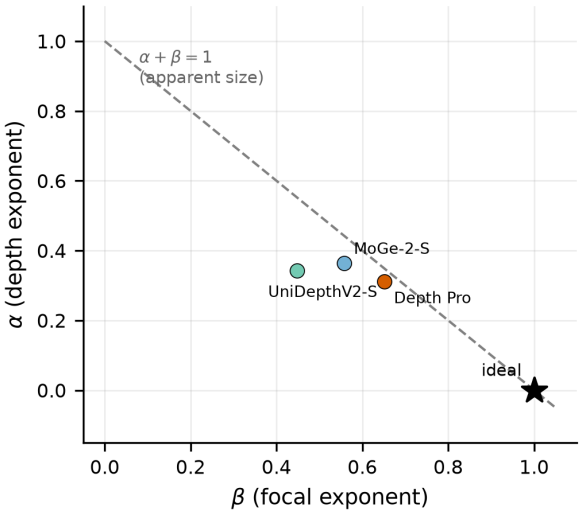


Figure 3. α–β plane. Dashed line: α+β = 1 (apparent size reasoning). Tara: ideal.

6. Result 2: label chhara score diye error anuman

|α| (label lage na) ar 3x zoom e AbsRel er Spearman ρ = 0.55 [0.18, 0.86], 32 ta model×dataset jora. Dataset er bhitore KITTI 0.98, NYU 0.95, DIODE-in 0.88, kintu DIODE-out -0.38. Shadharon (1x) accuracy predict kore na (ρ = -0.09). **Shotti kotha:** zoom e depth shrink korle error bare-i, tai sambandha kichuta nijer theke ashe. Claim ta tai choto: "zoom e kon model koto bhul korbe, label chhara bola jay."

7. Result 3: label chhara fine-tuning (EFT ar EFT+B)

Frozen copy (teacher) zoom chhara chhobi te ja bole, student ke zoom kora chhobi te ta-i bolte shekhai. Kono depth label ba camera info lage na. Training data: 7 ta KITTI raw drive (test drive er sathe overlap nai, code nijei check kore) ar NYUv2 er 795 ta train chhobi. DIODE kokhono training e dekhano hoy na.

Notun finding, resolution shortcut: shudhu zoom loss diye train korle (EFT) model shikhe ney "jhapsa chhobi = zoom". Blur control e KITTI te depth 30% bodle jay. EFT+B te ekta blur-invariance term jog kori: jhapsa (kintu zoom na) chhobite ager depth-i bolte hobe. Eta Zhong et al. (ICCV 2023) e nai.

Dataset	Model	α	AbsRel 1x	AbsRel 3x	Blur shortcut
KITTI	DA-V2-S (frozen)	1.03	0.138	0.665	1.1%
KITTI	EFT	0.18	0.156	0.158	29.6%
KITTI	EFT+B (ours)	0.05	0.175	0.178	4.9%
NYUv2	DA-V2-S (frozen)	0.98	0.344	0.563	0.7%
NYUv2	EFT	0.12	0.313	0.226	4.2%
NYUv2	EFT+B (ours)	0.07	0.301	0.266	0.4%
DIODE-in (unseen)	DA-V2-S (frozen)	0.70	0.280	0.558	1.0%
DIODE-in (unseen)	EFT	0.39	0.368	0.364	0.6%
DIODE-in (unseen)	EFT+B (ours)	0.37	0.400	0.367	0.4%
DIODE-out (unseen)	DA-V2-S (frozen)	0.65	0.731	0.551	0.6%
DIODE-out (unseen)	EFT	0.39	0.877	0.648	1.5%
DIODE-out (unseen)	EFT+B (ours)	0.40	1.000	0.676	0.4%

- EFT+B prai puro equivariant (KITTI ar NYU te α 0.05–0.07), ar blur shortcut prai chole geche.
- Trade-off: shadharon (1x) chhobite accuracy kichuta kome, beshi kome DIODE e (training e dekhe nai). λ = 3 (shokto anchor) diye eta kombe kina, sheta ekhon Kaggle e test hocche.
- "Anchor-only" baseline asole frozen model (student = teacher, tai gradient 0); er bodole blur-only baseline cholche.

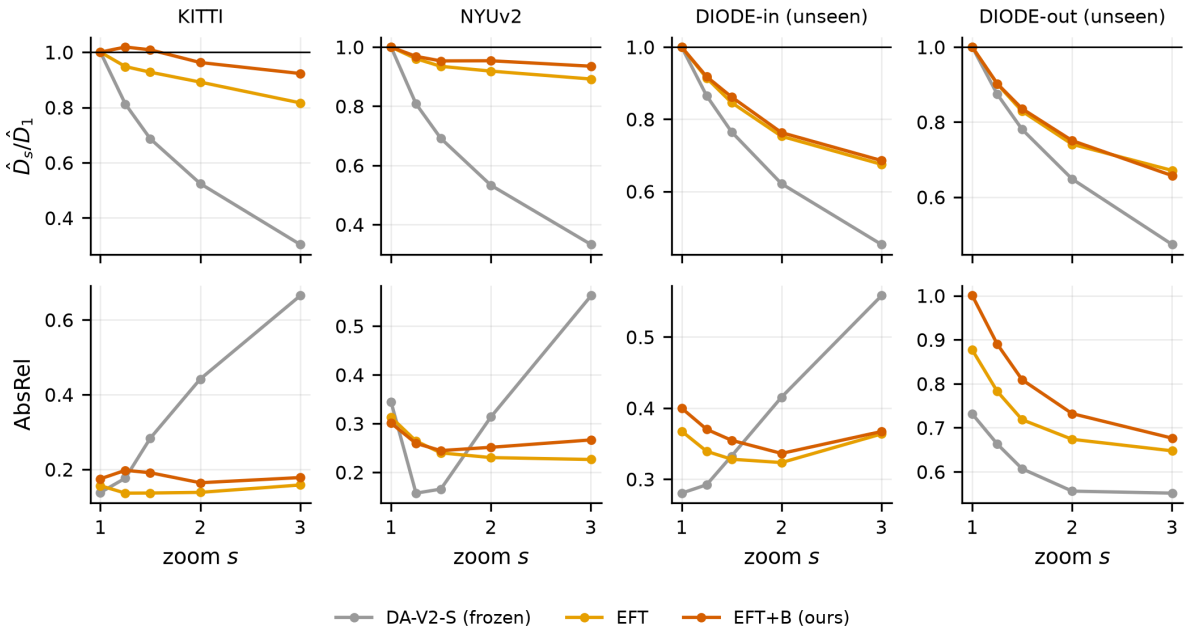


Figure 5. Upore: zoom e depth ratio (1 = ideal). Niche: AbsRel. Dhushor: frozen, holud: EFT, lal: EFT+B. Note: alada zoom e AbsRel alada pixel e mapa; NYU te frozen model 1.25–1.5x e 1x er cheye bhalo AbsRel pay, karon 1x e depth beshi bole ar zoom shrink kakatale thik er kachhe ane. Tai AbsRel na, α dekhte hobe.

8. Prathomik: model boro hole ki thik hoy? (1 ta chhobi, final na)

Model	3x e depth ratio (thik = 1)	3x e focal ratio (thik = 3)
DA3-L	– (relative depth)	1.55
DA3-Metric-L (thik K deya)	1.07	–
DA3-Nested	0.38	1.18
DA-V2-L	0.45	–
MoGe-2-B	0.66	2.07
MoGe-2-L	0.71	2.17
UniDepthV2-B	0.55	1.61
UniDepthV2-L	0.58	1.73
UniK3D-L	0.59	2.07
VGGT-1B	– (relative depth)	2.50

Shobcheye notun ar boro DA3-Nested (≈1.4B) 3x zoom e depth 0.38 bole ar focal prai bodlay na. Mane boro ar notun model holei zoom bujhe na. Eta shudhu 1 ta KITTI chhobi; puro run (100 chhobi × 4 dataset) Kaggle e cholche.

9. Shotti negative result

- Theory er "soft" prediction (scene e depth beshi chhoriye thakle α kombe) data te pawa jay nai (camera model e ρ ≈ 0).
- Label-free score shadharon accuracy predict kore na, shudhu zoom e accuracy.
- UniDepthV2-S DIODE-outdoor e lomba error tail dey (prai 10% chhobite 6x beshi depth); α er upor eta prabhab felbe na.
- Stale-K result notun na (Kim & Lee 2026 er kaj ke support kore).

10. Agerkar kaj ar amader notun ongsho

Paper	Ki koreche	Amader sathe tofat
Zhong et al., ICCV 2023	crop-resize equivariance, self-consistency loss	camera/metric nai; amra α, β, theory, notun model, shortcut + EFT+B
Kim & Lee, Electronics 2026	stale intrinsics e camera-input model er bhul	amader stale-K tader support kore
Roll-robust depth (2026)	camera roll	roll e intrinsics bodlay na, zoom e bodlay

Paper	Ki koreche	Amader sathe tofat
KPE (2023), PlaneDepth (CVPR 2023)	crop er perspective ambiguity	test/training input; zoom audit na
FS-Depth (2023)	crop diye focal augment, label shoho	amader label ar focal input chhara
DepthCues (CVPR 2025)	depth cue probing (size shoho)	amra size cue er physical meaning (α)

Amader notun ongsho: (1) metric depth er zoom audit with α ar β; (2) α+β = 1 line ar zoom–squash theory; (3) teen dhoroner model (pixels only / predicts camera / takes camera) ek scale e; (4) label-free proxy; (5) resolution shortcut finding ar EFT+B; (6) principal point test ar ZoomFix (EXIF zoom factor diye training chhara fix); (7) model ladder (DA3, VGGT, UniK3D shoho).

11. Code

- GitHub (private): github.com/rabeyanoor/zoom-equivariance
- 25 ta unit test (zoom geometry, audit toy model, analysis, ZoomFix), ruff lint, GitHub Actions CI
- Vectorised analysis (loop er cheye 150 gun druto, test e hubohu ek)
- verify_results.py: protita CSV te shob zoom, duplicate nai, shob model ekoi chhobi
- Paper er shob number make_paper_numbers.py diye CSV theke, haate kichu lekha hoy na
- Protita Kaggle run er log logs/kaggle/ e

12. Baki kaj

Kaj	Obostha
audit-a2: final 7 model (same-pixel reference, principal point, ZoomFix column)	Kaggle e cholche
Model ladder c1 (DA-V2-L, MoGe-2-B/L, UniDepthV2-B/L, UniK3D-L)	Kaggle e cholche
Model ladder c2 (DA3 ×3, Metric3D-L, VGGT)	GPU slot er opekkha
EFT ablation: blur-only, EFT+B λ = 3	GPU slot er opekkha
Abstract, intro, results, conclusion asol number diye	result ashle
(Optional) phone e 1x/2x/5x zoom chhobi, tape diye mapa dure	tumi korbe
Overleaf e PDF, supervisor review, CVPR submit (~7–16 Nov)	shesh e